Pranjal Arya

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Skills

- **Languages:** C, ARM Assembly, Bash scripting.
- **Protocols / Device drivers:** USB, SD Card, SDIO, UART, SPI, I2C and GPIO.
- **Processor:** ARM Cortex - (M4, M33).
- **Tools:** Saleae analyzer, SEGGER SystemView.
- **Miscellaneous:** Git, GDB, FreeRTOS, Linux System, TrustZone for Cortex-M, Pre-silicon and post-silicon bring-up.

Experience (3 years)

Member of the team responsible for firmware, device drivers, BSP, and middleware development for a family of Infineon PSoC devices.

Infineon Technologies | **Senior Software Engineer** | **April 2023 - Present**

- **Firmware development & bring-up**:
  - Performed Pre- and post-silicon bring-up on various ARM Cortex-M4 and M33 based MCUs.
  - Developed firmware and BSP for upcoming PSoC6 MCUs using different toolchains.
  - Ensured smooth integration of delivered assets to middlewares, RTOS.
  - Participated in code review and helped new team members ramp up.

Infineon Technologies | **Software Engineer** | **July 2020 - March 2023**

- **Device drivers**:
  - Designed and developed features in USB 2.0, SD Card, SDIO, UART, SPI, I2C, and GPIO device drivers.
  - Analyzed critical issues reported by customers and internal teams and provided fixes.
- **Middleware**:
  - Involved in USB - HID middleware development.
  - Upgraded PSoC6 hardware acceleration to support the latest MbedTLS release on public GitHub.
- **BSP and CI**: Created scripts and makefiles to automate build and basic sanity testing. Integrated this framework in CI using Git.

Cypress Semiconductor | **Intern** | **January 2020 - June 2020**

- **Internship project**: Unified device drivers to generate a common library to enable a family of PSoC6 devices from a single source.

Education

B.Tech. (Electronics & Communication Engineering) - Nirma University, Ahmedabad — CPI: **8.24** | 2016 - 2020
HSC — **88.9%** | 2016
SSC — **89.7%** | 2014

Projects

- **Custom Real-Time Operating System development from scratch** — GitHub project link
  Developed a custom RTOS kernel and added support for Cortex-M architecture with the following features:
  - Round-Robin scheduler
  - Heap memory manager supporting aligned memory allocation and reporting maximum memory usage statistics
  - Spinlock and semaphore support for synchronization
  - Mailbox for communication
- **Custom bootloader** — GitHub project link
  Developed a custom bootloader for STM32L476RG MCU with the following features:
  - Host application in PC interacts with a custom bootloader over UART.
  - Can maintain version information and update application firmware using an A/B swap mechanism.
- **CPU Faults on Arm Cortex-M** — GitHub project link
  To understand ARM processor internals, generated various Arm Cortex-M4 system exceptions using software on TM4C123GXL Tiva C series MCU. Wrote a fault handler to dump the status of the processor over debug prints.

Awards

- **YIP Award (Infineon Idea Management Program)** — Infineon Technologies, February 2023
  Awarded to a team of 2 members for internal framework improvement resulting in a 40% cost reduction.

- **Global Spot Award** — Infineon Technologies, January 2023
  Awarded for significant contribution to device driver library development.